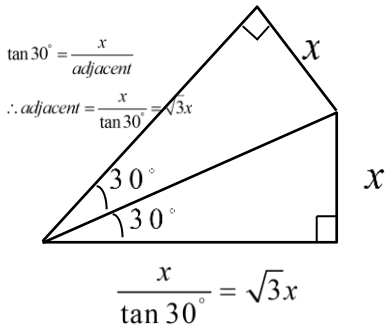


2013 A Level Paper 2
Section A: Pure Mathematics [40 marks]

Qn 1	Suggested Solutions
(i)	Since $R_g = \mathbb{R} \not\subset D_f = \mathbb{R} \setminus \{1\}$, fg does not exist.
(ii)	$gf(x) = 1 - 2\left(\frac{2+x}{1-x}\right) = 3\left(\frac{1+x}{x-1}\right)$ $\therefore gf : x \mapsto \frac{3(x+1)}{x-1}, x \neq 1.$ Note: $D_{gf} = D_f$ Let $(gf)^{-1}(5) = x$. $5 = gf(x) = \frac{3(x+1)}{x-1}.$ $x = 4.$ Note: $h^{-1}(y) = x \Rightarrow y = h(x)$

Qn2	Suggested Solutions
(i)	$V = \text{Cross Sectional Area} \times \text{Height}$ $= \frac{1}{2}(a - 2\sqrt{3}x)^2 \sin 60^\circ x$ $= \frac{x}{2}(a - 2\sqrt{3}x)^2 \frac{\sqrt{3}}{2}$ $= \frac{x\sqrt{3}}{4}(a - 2\sqrt{3}x)^2 \text{ (shown)}$  $\tan 30^\circ = \frac{x}{\text{adjacent}}$ $\therefore \text{adjacent} = \frac{x}{\tan 30^\circ} = \sqrt{3}x$ $\frac{x}{\tan 30^\circ} = \sqrt{3}x$
(ii)	$V = \frac{1}{4}x\sqrt{3}(a - 2x\sqrt{3})^2 = \frac{\sqrt{3}}{4}[x(a - 2x\sqrt{3})^2]$ $\frac{dV}{dx} = \frac{\sqrt{3}}{4}[(a - 2\sqrt{3}x)^2 + x \cdot 2(a - 2\sqrt{3}x)(-2\sqrt{3})]$ $= \frac{\sqrt{3}}{4}(a - 2\sqrt{3}x)[(a - 2\sqrt{3}x) - 4\sqrt{3}x]$ $= \frac{\sqrt{3}}{4}(a - 2\sqrt{3}x)[a - 6\sqrt{3}x] \text{-----(1)}$ When $\frac{dV}{dx} = 0$, $a = 2\sqrt{3}x$ or $a = 6\sqrt{3}x$. (rej since $a > 2\sqrt{3}x$ or else $V < 0$) $\therefore x = \frac{a}{6\sqrt{3}} = \frac{a\sqrt{3}}{18}$ Use Product Rule and Factorise common factors to simplify. DO NOT EXPAND!

Using 2nd Derivative Test

$$\text{From (1), } \frac{d^2V}{dx^2} = \frac{\sqrt{3}}{4} \left[-2\sqrt{3}(a - 6\sqrt{3}x) + (a - 2\sqrt{3}x)(-6\sqrt{3}) \right]$$

$$= -\frac{3}{2} \left[(a - 6\sqrt{3}x) + 3(a - 2\sqrt{3}x) \right]$$

$$= -\frac{3}{2} \left[a - 6\sqrt{3}x + 3a - 6\sqrt{3}x \right]$$

$$= -\frac{3}{2} \left[4a - 12\sqrt{3}x \right]$$

$$= -6(a - 3\sqrt{3}x)$$

$$\text{When } x = \frac{a\sqrt{3}}{6}, \frac{d^2V}{dx^2} = -6 \left(a - 3\sqrt{3} \cdot a \cdot \frac{\sqrt{3}}{6} \right) = 3a > 0. \Rightarrow V \text{ is min.}$$

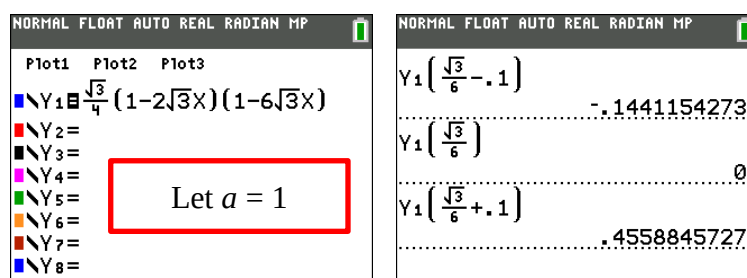
$$\text{When } x = \frac{a\sqrt{3}}{18}, \frac{d^2V}{dx^2} = -6 \left(a - 3\sqrt{3} \cdot a \cdot \frac{\sqrt{3}}{18} \right) = -3a < 0. \Rightarrow V \text{ is max.}$$

Alternative: Using 1st Derivative Test

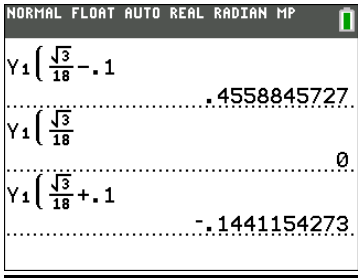
$$\frac{dV}{dx} = \frac{\sqrt{3}}{4} (a - 2\sqrt{3}x) [a - 6\sqrt{3}x]$$

x	$\left(\frac{a\sqrt{3}}{6} \right)^-$	$\frac{a\sqrt{3}}{6}$	$\left(\frac{a\sqrt{3}}{6} \right)^+$
$\frac{dV}{dx}$	-ve ↘	0 —	+ve ↗

$$\therefore V \text{ is min at } x = \frac{a\sqrt{3}}{6}.$$



x	$\left(\frac{a\sqrt{3}}{18} \right)^-$	$\frac{a\sqrt{3}}{18}$	$\left(\frac{a\sqrt{3}}{18} \right)^+$
$\frac{dV}{dx}$	+ve ↗	0 —	-ve ↘

	$\therefore V \text{ is max at } x = \frac{a\sqrt{3}}{18}.$  Calculator screenshot showing values for V at different x values: $V_1\left(\frac{\sqrt{3}}{18} - .1\right) = .4558845727$ $V_1\left(\frac{\sqrt{3}}{18}\right) = 0$ $V_1\left(\frac{\sqrt{3}}{18} + .1\right) = -.1441154273$ $\begin{aligned} \text{Max } V &= \frac{\sqrt{3}}{4} \left(\frac{a\sqrt{3}}{18} \right) \left[a - 2\sqrt{3} \left(\frac{a\sqrt{3}}{18} \right) \right]^2 \\ &= \frac{a}{24} \left(\frac{2a}{3} \right)^2 \\ &= \frac{a^3}{54} \end{aligned}$
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Qn 3 Suggested Solutions	
(i)	Let $y = f(x) = \ln(1 + 2\sin x)$. $e^y = 1 + 2\sin x$ $\frac{dy}{dx} e^y = 2\cos x$ $\frac{d^2y}{dx^2} e^y + \left(\frac{dy}{dx}\right)^2 e^y = -2\sin x$ $\frac{d^3y}{dx^3} e^y + \left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right) e^y + 2\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right) e^y + \left(\frac{dy}{dx}\right)^3 e^y = -2\cos x$ When $x = 0$, $f(0) = 0$, $f'(0) = 2$, $f''(0) = -4$ and $f'''(0) = 14$. $\therefore f(x) = 2x - 2x^2 + \frac{7}{3}x^3 + \dots$
(ii)	Refer to MF26 for Maclaurin series of e^x and $\sin x$ then replace x by ax and replace x by nx respectively $e^{ax} \sin nx = \left(1 + ax + \frac{(ax)^2}{2!} + \dots\right) \left(nx - \frac{(nx)^3}{3!} + \dots\right)$ $= nx + anx^2 + \dots$ Comparing coefficients, $n = 2$, $a = -1$. Third non-zero term $= \frac{((-1)x)^2}{2!} (2x) - \frac{(2x)^3}{3!} = x^3 - \frac{4x^3}{3} = -\frac{1}{3}x^3$

Qn 4	Suggested Solutions
(i)	$\cos \theta = \frac{\begin{vmatrix} 2 & -6 \\ -2 & 3 \\ 1 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & -6 \\ -2 & 3 \\ 1 & 2 \end{vmatrix}} = \frac{-12 - 6 + 2}{\sqrt{4+4+1}\sqrt{36+9+4}} = \frac{16}{21}$ Question asks for acute angle, so remember to place a modulus ! $\theta = 40.4^\circ \text{ (1d.p.)}$
(ii)	$p_1: 2x - 2y + z = 1$ $p_2: -6x + 3y + 2z = -1$ Using GC, the line of intersection is $\mathbf{r} = \begin{pmatrix} -\frac{1}{6} \\ \frac{2}{3} \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} \frac{7}{6} \\ \frac{5}{3} \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$
(iii)	Let B denote the point $(0,0,1)$ on p_1 and C denote the point $(0,0,-\frac{1}{2})$ on p_2. $\frac{\left \overrightarrow{BA} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \right }{\sqrt{4+4+1}} = \frac{\left \overrightarrow{CA} \cdot \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix} \right }{\sqrt{36+9+4}}$ Case 1 $\frac{\overrightarrow{BA} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}}{\sqrt{4+4+1}} = \frac{\overrightarrow{CA} \cdot \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix}}{\sqrt{36+9+4}}$ $\frac{\begin{pmatrix} 4 \\ 3 \\ c-1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}}{3} = \frac{\begin{pmatrix} 4 \\ 3 \\ c+\frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix}}{7}$ $\frac{8-6+c-1}{3} = \frac{-24+9+2c+1}{7}$ $c = -49$

NORMAL FLOAT AUTO REAL DEGREE MP
 PLYSMLT2 APP
SIMULT EQN SOLVER MODE
 EQUATIONS 3 4 5 6 7 8 9 1
 UNKNOWN 2 3 4 5 6 7 8 9 1
 AUTO DEC
 NORMAL SCI ENG
 FLOAT 0 1 2 3 4 5 6 7 8 9

NORMAL FLOAT AUTO REAL DEGREE MP
 PLYSMLT2 APP
SYSTEM MATRIX (2 x 4)
 $\begin{bmatrix} 2 & -2 & 1 & 1 \\ -6 & 3 & 2 & -1 \end{bmatrix}$

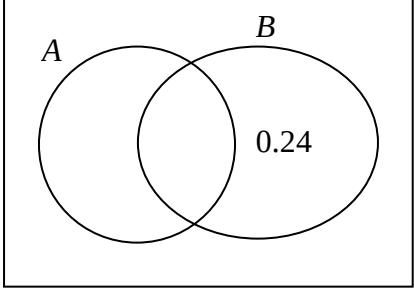
NORMAL FLOAT AUTO REAL DEGREE MP
 PLYSMLT2 APP
SOLUTION SET
 $x1 = -\frac{1}{6} + \frac{7}{6}x3$
 $x2 = -\frac{2}{3} + \frac{5}{3}x3$
 $x3 = x3$

	<u>Case 2</u> $\frac{\overrightarrow{BA} \cdot \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}}{\sqrt{4+4+1}} = \frac{\overrightarrow{CA} \cdot \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix}}{\sqrt{36+9+4}}$ $\frac{\begin{pmatrix} 4 \\ 3 \\ c-1 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}}{3} = \frac{\begin{pmatrix} 4 \\ 3 \\ c+\frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix}}{7}$ $\frac{-8+6-c+1}{3} = \frac{-24+9+2c+1}{7}$ $c = \frac{35}{13}$ Hence, the two possible values of c are -49 and $\frac{35}{13}$.
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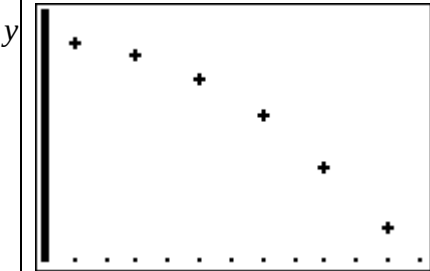
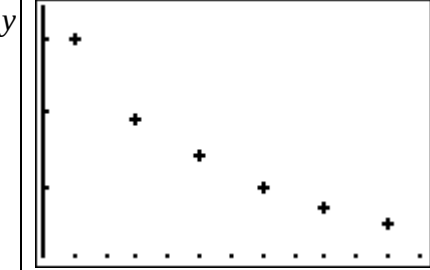
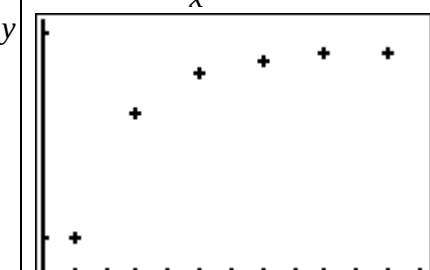
Qn 6	Suggested Solutions
	$P(Y < 2a) = 0.95$ $P\left(Z < \frac{2a - \mu}{\sigma}\right) = 0.95$ $\frac{2a - \mu}{\sigma} = 1.6449 \text{ -----(1)}$ $P(Y < a) = 0.25$ $P\left(Z < \frac{a - \mu}{\sigma}\right) = 0.25$ $\frac{a - \mu}{\sigma} = -0.67449 \text{ -----(2)}$ $(1) - (2): \frac{a}{\sigma} = 2.3194$ $\sigma = 0.43115a$ Substitute $\sigma = 0.43115a$ into (2): $a - \mu = -0.67449(0.43115a)$ $\mu = a + 0.67449(0.43115a)$ $= 1.29a \text{ (3 s.f.)}$

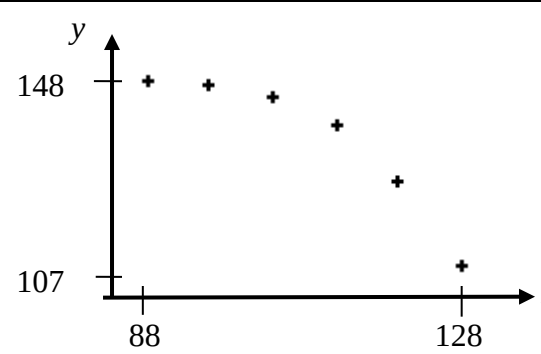
Note: The context must be stated with the assumptions !

Qn 7	Suggested Solutions
(i)	The probability of each packet of breakfast cereal containing a free gift is assumed constant (at 0.05). Whether a packet of breakfast cereal that contains a free gift is assumed to be independent of any other packets.
(ii)	Let F be the number of packets of breakfast cereal, out of n , that contains a free gift. $F \sim B\left(20, \frac{1}{20}\right)$ $P(F=1) = 0.37735$ $= 0.377 \text{ (to 3 s.f.)}$

Qn 8	Suggested Solutions
(i)	$P(B A') = 0.8$ $\Rightarrow \frac{P(B \cap A')}{P(A')} = 0.8$ $\Rightarrow P(B \cap A') = 0.8 \times (1 - 0.7)$ $\Rightarrow P(B \cap A') = 0.24$
(ii)	$P(A' \cap B') = 1 - [P(A) + P(B \cap A')]$ $P(A' \cap B') = 1 - [0.7 + 0.24]$ $P(A' \cap B') = 0.06$ 
(iii)	$P(A B') = 0.88$ $\Rightarrow \frac{P(A \cap B')}{P(B')} = 0.88$ $\Rightarrow \frac{P(A) - P(A \cap B)}{1 - P(B)} = 0.88$ $\Rightarrow \frac{P(A) - P(A \cap B)}{1 - [P(B \cap A') + P(A \cap B)]} = 0.88$ $\Rightarrow \frac{0.7 - P(A \cap B)}{1 - [0.24 + P(A \cap B)]} = 0.88$ $\Rightarrow 0.7 - P(A \cap B) = 0.88(0.76 - P(A \cap B))$ $\Rightarrow 0.12P(A \cap B) = 0.0312$ $\Rightarrow P(A \cap B) = 0.26$

Qn 9	Suggested Solutions																																													
(i)	Unbiased estimate of population mean, $\bar{x} = 12.8$ Unbiased estimate of population variance, $s^2 = 2.3057 = 2.31$ NORMAL FLOAT AUTO REAL RADIAN MP EDIT CALC TESTS 1:Edit... 2:SortA(3:SortD(4:ClrList 5:SetUpEditor NORMAL FLOAT AUTO REAL RADIAN MP <table><thead><tr><th>L1</th><th>L2</th><th>L3</th><th>L4</th><th>L5</th></tr></thead><tbody><tr><td>14</td><td></td><td></td><td></td><td></td></tr><tr><td>12.5</td><td></td><td></td><td></td><td></td></tr><tr><td>11</td><td></td><td></td><td></td><td></td></tr><tr><td>11</td><td></td><td></td><td></td><td></td></tr><tr><td>12.5</td><td></td><td></td><td></td><td></td></tr><tr><td>12.6</td><td></td><td></td><td></td><td></td></tr><tr><td>15.6</td><td></td><td></td><td></td><td></td></tr><tr><td>13.2</td><td></td><td></td><td></td><td></td></tr></tbody></table> NORMAL FLOAT AUTO REAL RADIAN MP EDIT CALC TESTS 1:1-Var Stats 2:2-Var Stats 3:Med-Med 4:LinReg(ax+b) 5:QuadReg 6:CubicReg NORMAL FLOAT AUTO REAL RADIAN MP 1-Var Stats List:L1 FreqList: Calculate NORMAL FLOAT AUTO REAL RADIAN MP 1-Var Stats $\bar{x}=12.8$ $\Sigma x=102.4$ $\Sigma x^2=1326.86$ $Sx=1.518457864$ $\sigma x=1.420387271$ $n=8$ $\min X=11$	L1	L2	L3	L4	L5	14					12.5					11					11					12.5					12.6					15.6					13.2				
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10	Solution
(i)	(A) $y = a + bx^2$, where a is positive and b is negative  (B) $y = c + d \ln x$, where c is positive and d is negative  (C) $y = e + \frac{f}{x}$, where e is positive and f is negative 

(ii)																																																																																					
(iii)	From the scatter diagram, as x increases, y decreases at an increasing rate, thus model (A) $y = a + bx^2$ is the most appropriate model. Using GC, $r = -0.939$ NORMAL FLOAT AUTO REAL RADIAN MP EDIT CALC TESTS 1:Edit... 2:SortA(3:SortD(4:ClrList 5:SetUpEditor NORMAL FLOAT AUTO REAL RADIAN MP <table border="1"> <thead> <tr> <th>L1</th> <th>L2</th> <th>L3</th> <th>L4</th> <th>L5</th> <th>3</th> </tr> </thead> <tbody> <tr><td>88</td><td>148</td><td></td><td></td><td></td><td></td></tr> <tr><td>96</td><td>147</td><td></td><td></td><td></td><td></td></tr> <tr><td>104</td><td>144</td><td></td><td></td><td></td><td></td></tr> <tr><td>112</td><td>138</td><td></td><td></td><td></td><td></td></tr> <tr><td>120</td><td>126</td><td></td><td></td><td></td><td></td></tr> <tr><td>128</td><td>107</td><td></td><td></td><td></td><td></td></tr> </tbody> </table> L3=L1² NORMAL FLOAT AUTO REAL RADIAN MP <table border="1"> <thead> <tr> <th>L1</th> <th>L2</th> <th>L3</th> <th>L4</th> <th>L5</th> <th>3</th> </tr> </thead> <tbody> <tr><td>88</td><td>148</td><td>7744</td><td></td><td></td><td></td></tr> <tr><td>96</td><td>147</td><td>9216</td><td></td><td></td><td></td></tr> <tr><td>104</td><td>144</td><td>10816</td><td></td><td></td><td></td></tr> <tr><td>112</td><td>138</td><td>12544</td><td></td><td></td><td></td></tr> <tr><td>120</td><td>126</td><td>14400</td><td></td><td></td><td></td></tr> <tr><td>128</td><td>107</td><td>16384</td><td></td><td></td><td></td></tr> </tbody> </table> NORMAL FLOAT AUTO REAL RADIAN MP EDIT CALC TESTS 1:1-Var Stats 2:2-Var Stats 3:Med-Med 4:LinReg(ax+b) 5:QuadReg 6:CubicReg 7:QuartReg 8:LinReg(a+bx) NORMAL FLOAT AUTO REAL RADIAN MP LinReg(a+bx) Xlist:L3 Ylist:L2 FreqList: Store RegEQ:Y1 Calculate NORMAL FLOAT AUTO REAL RADIAN MP LinReg y=a+bx a=189.7475283 b=-0.0046197847 r²=0.882101978 r=-0.9392028418	L1	L2	L3	L4	L5	3	88	148					96	147					104	144					112	138					120	126					128	107					L1	L2	L3	L4	L5	3	88	148	7744				96	147	9216				104	144	10816				112	138	12544				120	126	14400				128	107	16384			
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(iv)	Using GC, $y = 189.75 - 0.0046198x^2$ $\Rightarrow y = 190 - 0.00462x^2 \text{ (to 3 s.f.)}$ when $x = 110$, $y = 189.75 - 0.0046198(110)^2$ $= 134 \text{ (to 3 s.f.)}$																																																																																				

Qn 11	Suggested Solutions
(i)	Required Probability = $\frac{26 \times 25 \times 24 \times 9 \times 8}{26^3 \times 9^2} = \frac{1123200}{1423656} = \frac{400}{507} = 0.789$ Alternative: Required probability = $\frac{\binom{26}{3} \times 3! \times \binom{9}{2} \times 2!}{\binom{26}{1}^3 \times \binom{9}{1}^2} = \frac{1123200}{1423656} = \frac{400}{507} = 0.789$

(ii)	Required probability = $\frac{26^3 \times {}^9C_2}{26^3 \times 9^2} = \frac{4}{9}$ OR Case 1: Fix last digit as 9 Number of ways 26 × 26 × 26 × 8 × 1 Case 2: Fix last digit as 8 Number of ways 26 × 26 × 26 × 7 × 1 Total 8 cases Therefore Required Probability = $\frac{26^3 \times (8+7+6+\cdots+1)}{26^3 \times 9^2} = \frac{632736}{1423656} = \frac{4}{9}$	9C_2 to choose 2 different digits from 1-9 . Do not permute as the higher of these 2 digits will be at the second digit.
(iii)	Case 1: exactly two letter same, two digit different $\frac{\left({}^{26}C_1 \times {}^{25}C_1 \times \frac{3!}{2!} \right) \times \left({}^9C_2 \times 2! \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2} = \frac{140400}{1423656} = \frac{50}{507}$ Case 2: All letter same, two digit same $\frac{\left({}^{26}C_1 \right) \times \left({}^9C_1 \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2} = \frac{234}{1423656} = \frac{1}{6084}$ Case 3: All letter different, two digit same $\frac{\left({}^{26}C_3 \times 3! \right) \times \left({}^9C_1 \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2} = \frac{140400}{1423656} = \frac{50}{507}$ Therefore Required Probability = $\frac{50}{507} + \frac{1}{6084} + \frac{50}{507} = \frac{1201}{6084} = 0.197$	

	<u>Alternative:</u> Let A denotes the event “exactly two letters the same” Let B denotes the event “two digits the same” Therefore Required Probability = $P(A) + P(B) - 2P(A \cap B)$ $P(A) = \frac{\left({}^{26}C_1 \times {}^{25}C_1 \times \frac{3!}{2!} \right) \times \left({}^9C_1 \times {}^9C_1 \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2} = \frac{1950}{17576} = \frac{75}{676}$ $P(B) = \frac{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2} = \frac{1}{9}$ $P(A \cap B) = \frac{\left({}^{26}C_1 \times {}^{25}C_1 \times \frac{3!}{2!} \right) \times \left({}^9C_1 \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2} = \frac{17550}{1423656} = \frac{25}{2028}$ $\text{Required Probability} = \frac{75}{676} + \frac{1}{9} - 2 \left(\frac{25}{2028} \right) = \frac{1201}{6084} = 0.197$	Note: Question wants $P(A \cup B) - P(A \cap B)$ $= [P(A) + P(B) - P(A \cap B)] - P(A \cap B)$
(iv)	exactly one vowel (A, E, I, O or U) and exactly one even digit. Required probability $= \frac{\left({}^5C_1 \times {}^{21}C_1 \times {}^{21}C_1 \times \frac{3!}{2!} \times {}^4C_1 \times {}^5C_1 \times 2! \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2}$ OR $\frac{\left({}^5C_1 \times {}^{21}C_1 \times {}^{21}C_1 \times {}^3C_1 \times {}^2C_1 \times {}^4C_1 \times {}^5C_1 \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2}$ OR $\left(\frac{5}{26} \right) \left(\frac{21}{26} \right) \left(\frac{21}{26} \right) \frac{3!}{2!} \left(\frac{4}{9} \right) \left(\frac{5}{9} \right) 2!$ = 0.186	

	Alternative: Case 1: exactly one vowel and two same non-vowel and exactly one even digit. Case 2: exactly one vowel and two different non-vowel and exactly one even digit. Required Probability $= \frac{\left({}^5C_1 \times {}^{21}C_1 \times \frac{3!}{2!} \right) \times \left({}^4C_1 \times {}^5C_1 \times 2! \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2} + \frac{\left({}^5C_1 \times {}^{21}C_1 \times {}^{20}C_1 \times 3 \right) \times \left({}^4C_1 \times {}^5C_1 \times 2! \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2}$ = 0.186 Or $= \frac{\left({}^5C_1 \times {}^{21}C_1 \times \frac{3!}{2!} \right) \times \left({}^4C_1 \times {}^5C_1 \times 2! \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2} + \frac{\left({}^5C_1 \times {}^{21}C_2 \times 3! \right) \times \left({}^4C_1 \times {}^5C_1 \times 2! \right)}{\left({}^{26}C_1 \right)^3 \times \left({}^9C_1 \right)^2}$ = 0.186
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